
Deposit Insurance, Risk, and Market Power in Banking

Author(s): Michael C. Keeley

Source: *The American Economic Review*, Dec., 1990, Vol. 80, No. 5 (Dec., 1990), pp. 1183-1200

Published by: American Economic Association

Stable URL: <https://www.jstor.org/stable/2006769>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



JSTOR

is collaborating with JSTOR to digitize, preserve and extend access to *The American Economic Review*

Deposit Insurance, Risk, and Market Power in Banking

By MICHAEL C. KEELEY*

A fixed-rate deposit insurance system provides a moral hazard for excessive risk taking and is not viable absent regulation. Although the deposit insurance system appears to have worked remarkably well over most of its 50-year history, major problems began to appear in the early 1980's. This paper tests the hypothesis that increases in competition caused bank charter values to decline, which in turn caused banks to increase default risk through increases in asset risk and reductions in capital. (JEL 600)

It has long been recognized that a fixed-rate deposit insurance system, such as the Federal Deposit Insurance Corporation's (FDIC's), or the Federal Savings and Loan Insurance Corporation's (FSLIC's) can pose a moral hazard for excessive risk taking. The reason is that banks or thrifts can borrow at or below the risk-free rate by issuing insured deposits and then investing the proceeds in risky assets with higher expected yields.

As Robert C. Merton (1977) has shown, deposit insurance can be viewed as a put option on the value of a bank's assets at a strike price equal to the promised maturity value of its debt. Under a fixed-rate system, banks potentially can transfer wealth from the insuring agency, and, absent regulation, banks seeking to maximize the value of their equity will maximize the value of the put by

increasing asset risk and/or minimizing invested capital relative to assets.

Empirical research, however, does not seem to show that banks in general maximize the put option value. For one thing, many banks hold substantially more capital than the required amounts (Michael C. Keeley, 1988) and for another, researchers have found that for many banks, the value of the deposit insurance option is less than its price (Allan J. Marcus and Israel Shaked, 1984; Ehud Ronn and Avinash K. Verma, 1986; George Pennacchi, 1987), assuming that at the expiration of the option insolvent banks are closed. Moreover, for most of its 50-year history, the insurance system has been characterized by low failure rates and low payouts—just the opposite of what might be expected if banks were maximizing the value of the put option successfully.

Recently, bank and thrift failures and deposit insurance payouts have reached record highs (see Chart 1); the FSLIC has liabilities far in excess of its assets, and even the FDIC faces threats to its solvency. Although many have argued that these recent problems are in part due to the moral hazard of deposit insurance, the question is why it has taken 50 years for major problems to arise.

One explanation (Arnold Kling, 1986) is that the recent episode simply reflects an increasingly risky economy, which in turn has increased the risk of bank portfolios. In the last few years, whole sectors and regions of the national and even the world economy have encountered serious downturns that

*Vice President, Cornerstone Research, 1000 El Camino Real, Menlo Park, CA 94025. Much of the research in this paper was conducted while the author was a research officer at the Federal Reserve Bank of San Francisco. However, opinions expressed herein are those of the author and do not necessarily reflect the view of the Federal Reserve Bank of San Francisco, the Board of Governors of the Federal Reserve System, or Cornerstone Research. An earlier version of this paper was presented at Garn Institute of Finance's academic symposium on deposit insurance. Comments from William Beaver, Jack Beebe, Barbara Bennett, Mark Flannery, Christopher James, Ed Kane, Stuart Myers, Randall Pozdena, Anthony Saunders, and two anonymous referees are greatly appreciated. Alice Jacobson provided expert research assistance. The usual caveats apply, however.

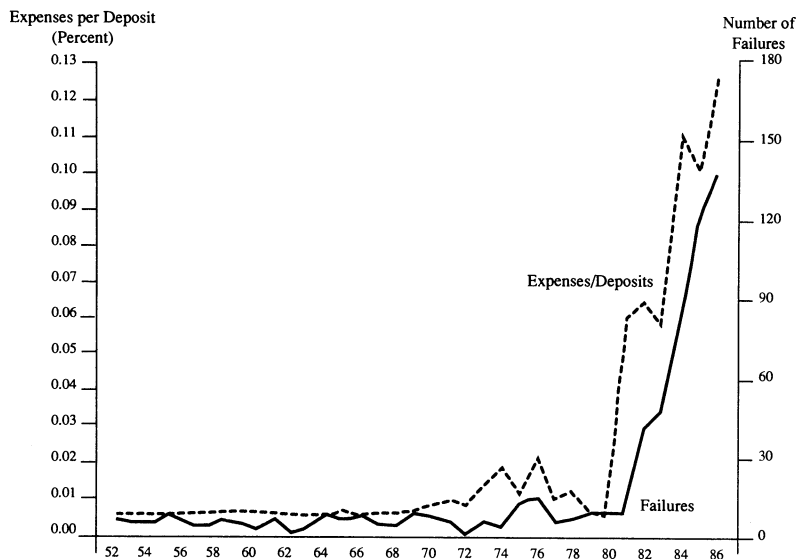


CHART 1. DEPOSIT INSURANCE EXPENSES PER DOLLAR OF DEPOSITS AND BANK FAILURES

have affected the values of bank and thrift assets. Similarly, interest rates have become more volatile, increasing the riskiness of banks', and especially thrifts', portfolios.

The rise in bank and thrift failures in recent years also may reflect the secular decline in capital-to-asset ratios over the past two decades. As Chart 2 shows, both market and book capital ratios of the 25 largest bank holding companies have fallen well below their levels in the mid-1950's, when only a handful of banks and thrifts failed each year, as opposed to several hundred per year recently. Moreover, beginning in about 1974, market values of the 25 largest bank holding companies in the aggregate fell below book capital ratios.

There are two reasons why declining capital ratios could lead to an increased rate of bank failures. First, lower capital, holding asset risk constant, leads to less protection against failure. Second, as shown in Frederick T. Furlong and Keeley (1987, 1989), lower capital ratios increase the incentive for banks to increase asset risk. Thus, even if overall risk in the economy did not increase, banks would have a greater incentive to increase asset portfolio risk due to the decline in capital ratios.

There is little doubt that increased risk in the economy and declining capital ratios have had a lot to do with the increase in bank and especially thrift failures in recent years. But these developments do not explain why banks and thrifts allowed bankruptcy risk to increase. After all, depository institutions have considerable control over the riskiness of their asset portfolios and perhaps even more control over their capital ratios. Thus, these explanations beg the question of why capital ratios behaved as they did.

Specifically, why did banks on average hold so much capital during the 1950's and early 1960's, and why did capital ratios fall during the 1960's and 1970's? It seems difficult to pinpoint any explicit regulatory changes that would have made it easier for banks to increase default risk, and banks had access to fixed-rate deposit insurance throughout the period.¹ A similar puzzle

¹Although the percentage of deposits explicitly covered by deposit insurance has increased, in theory, even partial deposit insurance coverage provides an incentive for banks to minimize capital and maximize asset risk (as long as they can share losses with the deposit insurance fund).

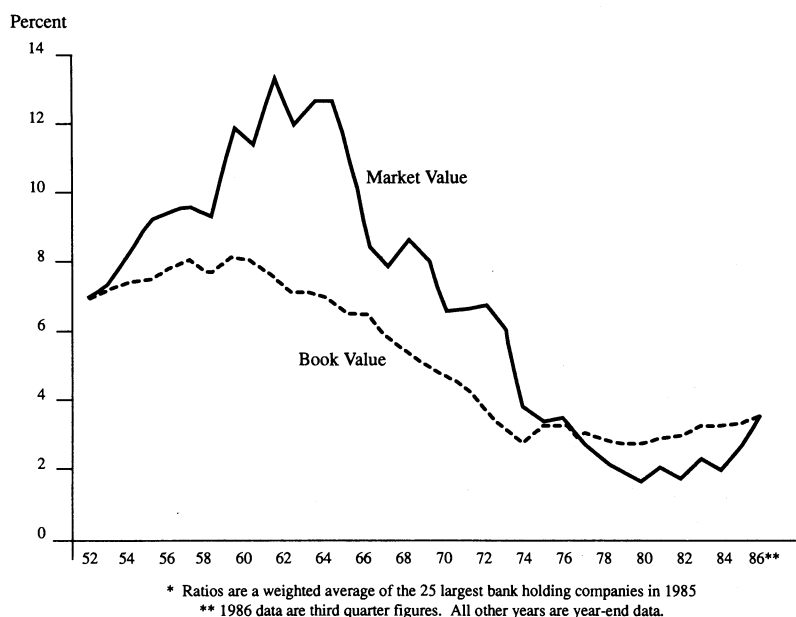


CHART 2. CAPITAL-TO-ASSET RATIOS, MARKET AND BOOK VALUES

arises in trying to explain the cross-sectional variation in bank capital ratios. Some banks, for example, hold much more capital than others and much more than regulators require.

This paper argues that one explanation for these apparent puzzles involves differences and changes in the degree of competition faced by banks. In the 1950's and even early 1960's banks partially were protected from competition by a variety of regulatory barriers. For example, chartering was very restrictive (Sam Peltzman, 1965) until the mid-1960's when James Saxton, then comptroller of the currency, greatly liberalized it (Keeley, 1985a, b). Moreover, some banks were protected by various state laws that limited or prohibited branching, multi-bank holding company, and interstate bank expansion. However, these laws have been greatly liberalized over the last few years, possibly eroding banks' charter values. Likewise, deposit rate deregulation may have diminished charter values by increasing competition, especially for institutions in protected local markets that had been relying on nonprice service competition to attract funds. In addition, beginning in the

early 1980's, thrifts were given expanded powers that enabled them to compete more fully with banks. Finally, many argue that changes in technology have increased the competition that banks face from nonbank financial firms, such as investment banks, brokerage firms, and insurance companies. Such developments as money market mutual funds, cash management accounts, and increased use of commercial paper have all made competitive inroads in banks' traditional product markets.

Increased competition may have reduced banks' incentives to act prudently with regard to risk taking. In fact, the evidence in Chart 2 of declining market values (which would reflect capitalized charter values) relative to book values (which would not) suggests that bank charter values were declining. In the 1950's and early 1960's, regulatory restrictions on entry and competition made bank charters valuable. With valuable charters as assets, banks had an incentive not to risk failure since the owners of the banks cannot sell the charter once the bank is declared insolvent. Instead, a bank that was insolvent on a book-value basis still had a valuable charter that the FDIC could sell

in a purchase and assumption (P&A). (In calculating primary book capital, intangible assets, generally representing the excess of the purchase price of assets that had been acquired over their book value, are subtracted from capital.) This may explain why P&A's typically are less costly than liquidations. Also, banks would apparently be willing to overpay for deposit insurance if it were needed to obtain and maintain a valuable charter.

The possession of a valuable charter thus made it difficult for banks to shift losses to the FDIC, and its potential loss in essence created a regulatory bankruptcy cost from the point of view of bank owners. This is especially so since regulators focus on book value when assessing a bank's solvency, not market value. Thus, the gains from feasible increases in risk taking would be offset by the diminished expected value of the charter. As a result, a bank will not have an incentive on the margin to increase default risk (either through reducing capital relative to assets or increasing asset risk) as long as the expected loss of the charter exceeds the gain to the bank of the enhanced value of the deposit insurance put option. Moreover, regulation limited the feasible increases in risk taking so as to prevent banks from potentially imposing losses that would have exceeded their charter values. This idea is established formally below using a state preference model.

In the empirical analysis below a simultaneous equations model of bank risk taking and charter value is developed and estimated. Changes in the laws governing branching, multibank holding company expansion, and interstate entry are used to identify the model statistically. Over the last 20 years or so, these anticompetitive laws have been liberalized greatly, and there are virtually no cases of states increasing their stringency (Dean F. Amel and Daniel G. Keane, 1987). Although the liberalization of these laws is not necessarily the most important factor in increasing the degree of bank competition, it is an easily observed exogenous factor with respect to bank risk taking. Thus, changes in these laws over time provide an opportunity to examine their

influence on market power in banks and whether exogenous variations in market power are related to bank risk taking.

This paper first examines the relationship between changes in regulatory entry barriers and the market power of banks in order to create an instrument for charter value. Then the relationship between market power and risk taking is estimated. This paper employs James Tobin's q , as suggested by Eric Lindenberg and Stephen Ross (1981), as a measure of a bank's market power (monopoly rents). Two measures of bank risk are then related to exogenous variations in q : the market-value capital-to-asset ratio and the interest cost on large, uninsured CD's. I find that q appears to be a useful proxy for market power and that banks with greater market power hold more capital and pay lower rates on CD's.

The remainder of the paper is organized as follows. In Section I, a state preference model is employed to show how market power can affect bank risk taking and how it can be measured. Section II presents the empirical results. Finally, Section III contains a summary and conclusions.

I. Theoretical Framework

Below, a state preference model is used to develop the major results.² The model described below closely follows that presented in Furlong and Keeley (1989). Marcus (1984) has developed similar results using an options model, but the state preference model clarifies the conditions under which a bank can benefit from increasing default risk and also illustrates the relationship between charter value and Tobin's q .

A two-period (current and future period), two-state model is used where P_1 and P_2 are the current values of a dollar payment in future states 1 and 2, respectively. (Thus, the risk-free interest rate is $1/(P_1 + P_2) - 1$).

²State preference models have been widely used in the analysis of banking and deposit insurance—see John H. Kareken and Neil Wallace (1978), William F. Sharpe (1978), Uri Dothan and Joseph Williams (1980), and Furlong and Keeley (1987, 1989).

The state prices are assumed to be exogenously given. To fund its assets, a bank uses an initial capital of C_0 dollars and issues deposits with a current value of D_0 dollars. Initially, the bank is assumed to issue risk-free deposits that pay off \$1 in each state, although this assumption is relaxed later to allow for the issuance of risky deposits. Also, it is assumed that initially the bank is not insured, although this assumption is relaxed later too. The bank invests in an asset security A that pays A_1 dollars in state 1 and A_2 dollars in state 2.

The current value of the bank's equity, V_0 , is found by valuing the various cash flows at the state prices. It is assumed that the bank can acquire security A at a price P_A and issue deposits at a price P_d . Thus, the bank can purchase $(C_0 + D_0)/P_A$ units of A. The current value of the bank's equity, V_0 , is the value of the cash flows from the assets acquired minus those of the liabilities issued:

$$(1) \quad V_0 = [(C_0 + D_0)/P_A][P_1 A_1 + P_2 A_2] - (D_0/P_d)(P_1 + P_2).$$

If the bank is competitive in both the asset market and the deposit market, then $P_A = P_1 A_1 + P_2 A_2$ and $P_d = P_1 + P_2$, and equation (1) above reduces to

$$(2) \quad V_0 = C_0.$$

If the bank is not insured, the value of its equity is independent of its (*ex ante*) risk taking. The reason is that if the bank were to default in state 1 and not meet its promised obligations to depositors, the depositors would demand sufficiently higher payments in state 2 so as to leave the costs unchanged at P_d .

However, with deposit insurance, depositors would not demand a higher payment in state 2, because the insuring agency would pay them the difference between the promised obligations and the asset value in the event of bankruptcy. But with fixed-rate underpriced deposit insurance with a premium, assumed for expositional purposes to be zero, banks pay less than the promised

amount if bankruptcy occurs in state 1, and only the promised amount in state 2. If the bank were to go bankrupt in state 1, then the future value of the bank's equity in state 1 is zero, and its current equity value is

$$(3) \quad V_0 = (C_0 + D_0)(P_2 A_2)/P_A - P_2 D_0/P_d > C_0.$$

The current value of the bank's equity when bankruptcy would occur in state 1 (that is, when bankruptcy is possible) equals the value of the excess of its deposit obligations over asset returns (the option value of deposit insurance) I_0 , plus its invested capital, C_0 . That is,

$$(4) \quad V_0 = I_0 + C_0$$

where

$$(5) \quad I_0 = D_0 P_1/P_d - (C_0 + D_0)P_1 A_1/P_A$$

and $I_0 > 0$.

As is well known, increasing capital, holding constant deposits, reduces the value of the deposit insurance option, and hence equity, and increasing asset risk (increasing the payment in state 2 while reducing that in state 1 so as to hold the price of the asset constant) increases the option value.³ Thus, the problem facing bank regulators is to constrain banks' incentives to reduce capital relative to assets and to increase asset risk. The puzzle is why regulators apparently succeeded throughout much of the last 50 years but in recent years apparently have failed.

³Since $I_0 > 0$, the value of the bank exceeds its initial capital investment. Thus,

$$dI_0/dC_0|D_0 = -P_1 A_1/P_A < 0$$

and

$$dI_0/dA_2|P_A = -(C_0 + D_0)(P_1/P_A)(dA_1/dA_2) > 0$$

since

$$dA_1/dA_2 < 0.$$

A. Charter Values

If banks can operate only with charters that are limited in supply, banks may be able to acquire assets at below-market prices (that is, bank loans would earn higher risk-adjusted rates than would market securities) and/or they may be able to make below-market-value payments on deposits (that is, deposits would pay below the risk-adjusted rate). Bank charters have been made valuable by limiting their supply and by protecting banks through various regulations that limited interbank competition as well as competition by nonbank firms.

In the model below, banks are assumed to be insured (at zero cost)⁴ but face periodic examinations. At the end of the period, if the bank is insolvent (that is, its assets, not including the charter value, are less than deposit obligations), equity holders receive nothing, depositors receive their promised obligations, and the insurance agency receives the bank's assets, including the charter. If the bank is solvent, however, the bank retains its charter value and continues to operate for another period.

If a bank chooses capital and asset risk so that it will not default in either state, the current value of the bank's equity is

$$(6) \quad V_0 = (C_0 + D_0)(P_1 A_1 + P_2 A_2) / P_A \\ - D_0(P_1 + P_2) / P_d + X_0$$

where

$$(7) \quad X_0 = P_1 X_1 + P_2 X_2$$

in which $P_1 X_1$ is the current value of the charter to operate one more period if state 1 occurs, and $P_2 X_2$ is the current value of the charter to operate one more period if state 2 occurs. Thus, the bank must balance the gains from increased risk taking (I_0) with the loss of the charter value if bankruptcy occurs ($P_1 X_1$). The bank will risk

⁴Similar results are obtained if deposit insurance has a positive cost not related to default risk. The assumption of zero cost is employed to simplify the analysis.

bankruptcy only if

$$(8) \quad I_0 > P_1 X_1.$$

Consider a bank just on the verge of insolvency in state 1 (that is, when a marginal increase in asset risk or reduction in capital would cause bankruptcy). The value of such a bank is

$$(9) \quad V_0 = (C_0 + D_0)(P_2 A_2) / P_A \\ - D_0 P_2 / P_d + X_0.$$

However,

$$(10) \quad dV_0 / dD_0 | C_0 = -P_1 X_1 < 0.$$

That is, a marginal increase in deposits holding capital constant, which causes bankruptcy in state 1, in turn causes the bank to lose the value of its charter if state 1 occurs, $P_1 X_1$. Similarly,

$$(11) \quad dV_0 / dA_2 | P_A, C_0 = -P_1 X_1 < 0.$$

Thus, a bank initially at a position where solvency is guaranteed in both states will not have an incentive on the margin to increase risk either through increases in leverage (that is, increases in deposits holding capital constant) or increases in asset risk.⁵ This remains true throughout the region where $P_1 X_1 > I_0$.

Although valuable bank charters do not obviate the need for bank regulation because I_0 is unbounded in the absence of

⁵Marcus (1984) shows that dV_0 / dD_0 can be positive for banks with low charter values and negative for banks with high charter values using an options pricing formula. Although he argues that $dV_0 / d\sigma$ (the equivalent of dV_0 / dA_2) can be negative, his figure 2 implies that, for banks with high charter values, $dV_0 / d\sigma$ is positive. Moreover, in his model, it is unclear what determines the critical value of whether a marginal increase in default risk will be beneficial.

In contrast, the state preference model shows that the choice is one of balancing the gains in the option value of deposit insurance with loss of the expected value of the charter. Marginal increases in default risk will not benefit the bank until default risk is sufficiently high so that $I_0 = P_1 X_1$.

regulation, they make the regulator's job much easier. If a bank's capital and asset risk were initially set so that solvency were assured in both states, a large discrete increase in asset risk or reduction in capital sufficient to make $I_0 > P_1 X_1$ would be required if the value of the bank's equity were to increase. Since such large discrete changes presumably would be easy to detect, banks would be discouraged from trying to increase default risk in the first place, and regulators would not need to be concerned with small changes in asset risk or capital.

B. Market Power

An uninsured bank that has market (monopoly) power in its asset market can make positive net present value loans. That is, the loans' future payoffs, when valued at the exogenously given state prices, exceed their current cost. For a such a bank,

$$(12) \quad (P_1 A_1 + P_2 A_2) / P_A = \varepsilon > 1.$$

However, the bank does not face an inexhaustible supply of such loan opportunities, and as assets A_0 (which equal $C_0 + D_0$) increase, ε , the ratio of cash flows from an asset to its price diminishes. (That is, $\varepsilon = \varepsilon(A_0)$ and $\varepsilon' < 0$.) Assuming such a bank maximizes its net-of-capital investment value, it will expand until the current value of its marginal revenue equals the marginal cost of its deposits (which is 1). That is, $\varepsilon(1+n) = 1$, where n is the elasticity of ε with respect to A_0 . (For an uninsured bank, this condition holds regardless of whether bankruptcy occurs, assuming no bankruptcy costs, since the costs of deposits are unaffected by the risk of bankruptcy.) Similar conditions hold for market power on the deposit side.⁶

For banks that have market power in either the asset or deposit market, as suggested originally by George Stigler (1964) and later and more formally by Lindenberg

⁶For a bank with market power on the deposit side, $(P_1 + P_2) / P_d = f < 1$.

and Ross (1981), Tobin's q is an ideal measure of market power. In this paper, Tobin's q is defined as the current market value of a firm's assets (the market value of its equity plus debt) divided by their current cost to a firm. The reason that q is an ideal measure of monopoly rents is that the capitalized value of such rents, whether they arise from market power in the asset market, deposit market, or both, will be reflected in the market value of the firm's equity, and thus assets, but not in the costs of acquired assets. The reasons for q 's superiority as a measure of market power are spelled out in more detail in Michael Salinger (1984) and Michael Smirlock et al. (1984). To see why Tobin's q is a measure of market power in the above model, note that q is given by

$$(13) \quad q = \frac{[(C_0 + D_0)\varepsilon - D_0 + X_0] + D_0}{C_0 + D_0} \\ = \varepsilon + X_0 / (C_0 + D_0).$$

(The terms in brackets [] represent the market value of the bank's equity to which the current value of debt D_0 is added.) Equation (13) shows that q is equal to the current plus future degree of market power as reflected in current and future ε . For a competitive firm, $\varepsilon = 1$ and $X_0 = 0$, but for a firm with market power, as discussed above, $\varepsilon > 1$ and $X_0 > 0$. Thus, an uninsured bank with no market power in either the asset or deposit market would have a q of 1.⁷

II. Empirical Evidence

The data used to estimate the model are from several sources. The bank holding company data are from the Compustat bank

⁷For a bank with market power on the deposit side as well,

$$q = \varepsilon + \frac{D_0}{C_0 + D_0}(1-f) + \frac{X_0}{C_0 + D_0} \quad \text{where } f < 1$$

is inversely related to the degree of market power on the deposit side (see footnote 6). Thus, a bank with market power on the deposit side also will have a q greater than one.

tapes, which contain balance sheets, income statements, and monthly stock prices for the 150 largest bank holding companies (BHC's). Although this sample is not representative of the entire population of all banks or bank holding companies, which comprises many smaller and often privately held organizations, the BHC's in this sample hold about 40 percent of all bank assets and thus are of interest in their own right. Data on the interest cost of large, uninsured CD's are from the Bank Consolidated Report of Conditions and Income (the Bank Call Report). Data on state branching, multibank holding company expansion, and interstate entry laws are from Amel and Keane (1987) and various Federal Reserve Annual Statistical Digests.

A. Measuring Market Power

As discussed above, q is used to measure the degree of market power in banking. The measure q is defined as the market value of assets (calculated as the sum of the market value of common equity—price per share times number of shares—and the book value of liabilities) divided by the book value of assets.⁸ The assumption is that the capitalized value of the bank charter will be reflected in the market value of equity (and thus the market value of assets as defined above), but not the book value of equity or assets.⁹ Thus, banks with larger charter val-

ues due to market power in asset and/or deposit markets should have greater market-to-book asset ratios. Note that the ability to issue deposits at below-market rates is an asset, the value of which will be reflected in the market value of the bank's equity and thus the market value of assets as I define them.

Several difficulties arise in using q as a measure of a bank's market power. First, the book value of assets represents the historical costs of assets acquired and sold over time, not the current costs of the assets. Thus, *ex post* market-to-book ratios that differ from 1 may reflect different asset return realizations rather than the degree of market power, which would be reflected in the *ex ante* q . Thus, the theoretically appropriate *ex ante* q is measured with error when using the *ex post* q . Another reason that q may not accurately reflect the degree of market power is that the value of potentially underpriced deposit insurance could be capitalized into a bank's market value.¹⁰ Several methods are used in the empirical analysis to control for these possibilities.

First, and most importantly, simultaneous equations techniques are used. In the first stage, an instrument is created for q , and then the predicted ratio is used as an explanatory variable in second-stage equations of bank risk taking. Thus, the empirical model allows for both the possible endogeneity of q and the fact that q is measured with error.

Second, a sample of banking organizations is selected to have similar histories. To do this, the sample is restricted to banking organizations for which data are continuously available from 1970 through 1986. Moreover, two variables are included to control for different asset histories: a dummy variable for banks that were on the Computat tapes in 1964 and survived to be included in the sample (about 38 percent of

⁸Banks use an accounting convention in which loan loss reserves are counted as book capital. However, since loan loss reserves are often taken only when asset losses either have already occurred or when they are anticipated, they in fact often do not represent book capital since the addition to capital usually would be offset at least fully by asset losses if they were realized on the books. Thus, in this paper, loan loss reserves are not counted as book capital. (This procedure follows generally accepted accounting procedures as followed by bank holding companies in their 10-K and 10-Q reports filed with the Securities and Exchange Commission.)

⁹Although a firm that acquired a bank originally endowed with a valuable charter would record that charter's value (i.e., the excess of the purchase price over the book value) as an intangible asset, intangible assets are not counted as primary book equity, the measure of book equity used in this paper.

¹⁰As Smirlock and Gilligan (1984) argue, a q greater than 1 also could reflect the capitalized value of a firm-specific efficiency enhancing factor of production. However, a firm possessing such a factor would have the same incentives to protect its value as would a firm possessing market power.

the sample) and each bank's asset growth rate since 1970.

B. Model Structure

The empirical model consists of two sets of equations. In the first set of regressions, q is regressed on dummy variables that equal 1 during a given period if there was a liberalization in the laws governing state branching, multibank holding company, and interstate expansion, respectively, during any previous period. These regressions also contain a set of control variables and other proxies for market power such as the ratio of demand deposits to total deposits, the ratio of foreign deposits to total deposits, and the ratio of loans to assets. (The branching variables are constructed based on data in Amel and Keane [1987] and various Federal Reserve Statistical Abstracts.) The balance sheet data are from Compustat and refer to the consolidated holding company.

The hypotheses are that unanticipated liberalization of legal entry barriers should erode banks' market power and thus negatively affect q and that greater deposit funding and loan making also might be related to market power and thus positively affect q .

In a second set of regressions, both actual q and an instrument for q are explanatory variables in equations that attempt to explain bank risk taking. The hypotheses are that the banks with greater market power should have larger capital-to-asset ratios and lower risks of default.

The system to be estimated can be represented as

$$(14) \quad q_{it} = X_{1it}\beta_1 + \epsilon_{1it}$$

$$(15) \quad \text{risk}_{it} = X_{2t}\beta_2 + q_{it}\beta_3 + \epsilon_{2it}$$

where

X_{1it} is a vector of branching, financial, control, and other variables for bank i at time t ;

X_{2t} is a vector of financial and other control variables at time t ;

q_{it} is the bank's market-to-book asset ratio at time t [in the two-stage least-squares estimates, equation (14) is used to construct an instrument for q_{it}];

risk_{it} is a measure of bank default risk (two measures are used: the market-value capital-to-asset ratio [the market value of common equity divided by the market value of equity plus the book value of liabilities] and the interest cost on large CD's);

β_1 and β_2 are vectors of coefficients;

β_3 is the effect of q on risk; and

ϵ_{1it} and ϵ_{2it} are random error terms.

C. Empirical Results

Table 1 contains descriptive summary statistics for the bank holding companies in the sample using fourth-quarter (year-end) data for the 1970–86 period.¹¹ It is interesting to note that market-to-book asset ratios range from 0.95 to 1.18 and that market-value capital-to-asset ratios range from 0.0075 to 0.21. Although it is not shown in the table, about 32 percent of the bank holding companies were in states that liberalized branching laws, 44 percent were in states that liberalized multibank holding company expansion laws, and 78 percent were in states that liberalized interstate entry laws by 1986.

D. Market-to-Book Asset Ratios

Table 2 contains the results of the estimates of equation (14) over the 1971–86 period using fourth-quarter data. Estimates of the effects of four different types of variables are reported: branching variables, control variables, balance sheet variables, and financial variables. The branching variables are dummies reflecting the liberalization during a previous period of branching,

¹¹Some of the data items were missing on the Compustat tapes for particular bank holding companies in particular quarters. Rather than exclude the entire observation, missing values were forecast on the basis of quadratic time-trend OLS regressions estimated separately for each bank. For variables known a priori to be nonnegative, forecast values were constrained to be nonnegative.

TABLE 1—SUMMARY STATISTICS FOR 85 LARGE BANK HOLDING COMPANIES
(POOLED 1970–86, FOURTH-QUARTER DATA)

Characteristic	Mean	Minimum	Maximum
Liberalization of Branching Law (Dummy Variable)	0.19	0	1
Liberalization of Multibank Holding Co. Law (Dummy Variable)	0.26	0	1
Liberalization of Interstate Entry Law (Dummy Variable)	0.12	0	1
Book Value of Assets (Net of Loan Loss Reserves in \$ Millions)	\$10,587	\$278	\$195,147
Market-to-Book Asset Ratio, q	1.00	0.95	1.18
Multinational Regulatory Status (Dummy Variable)	0.19	0	1
Foreign Deposits/Total Deposits	14 percent	0 percent	89 percent
(Cash + Treasury Securities)/Total Assets	24 percent	7.8 percent	50.1 percent
Annual Assets Growth Rate Since 1970	12 percent	–1 percent	63 percent
Demand Deposits/Total Deposits	35.9 percent	0 percent	72.3 percent
Market-Value Capital-to-Asset Ratio	0.056	0.0075	0.21
Book-Value Capital-to-Asset Ratio	0.055	0.010	0.14
New York Composite Index	70.8	35.4	142.1
3-Month Treasury-Bill Rate	7.64 percent	4.02 percent	15.66 percent
20-Year Treasury-Bond Rate	9.03 percent	5.96 percent	13.73 percent
Average Maturity of CD's (Months)	6.45	1.19	19.81
Interest Cost of CD's	0.085	0.050	0.13

multibank holding company, and interstate expansion laws in the state in which the bank is located. The financial variables are the New York Composite Index, the three-month Treasury-bill rate, and the 20-year Treasury-bond rate. They are included to control for the effects of general interest rate and stock market trends on the market-to-book ratio that would not be related to changes in market power.

Other variables are included as proxies for market power. While some of them might be endogenous, the coefficient estimates of branching variables are not sensitive to their inclusion. The estimated coefficients generally conform with a priori expectations. Liberalization of branching or multibank holding company expansion laws in a previous period is associated with a statistically significantly (at the 1-percent level) lower market-to-book asset ratio. This suggests that both branching and multibank holding company expansion restrictions do provide banks a degree of protection from competition. This finding is consistent with a study by Mark J. Flannery (1984) that finds that unit banks in unit banking states

earn monopoly profits approximately 20 percent above those reported by similar banks in branching states, as well as other studies that have found that branching and multibank holding company expansion restrictions lead to higher loan rates and lower deposit rates.¹² (See Allen N. Berger and Timothy H. Hannan [1987] for recent evidence that restricted branching leads to lower deposit rates.) However, no significant effect of liberalization of interstate entry restrictions is found. This may reflect the fact that these laws generally allow entry only by acquisition (which would increase market prices) and do not allow *de novo* entry, which would directly increase competition and thus diminish market prices.

¹²There is an extensive literature on the effects of branching restrictions on competition. Generally, these studies show that branching restrictions are associated with reduced competition. For example, Donald T. Savage and Stephen A. Rhoades (1979) found banks in statewide branching states paid higher rates on deposits. Also, a survey by George J. Benston (1973) finds general gains in service for the banking public. See Savage and Elinor H. Solomon (1980) for a discussion of the literature.

TABLE 2—POOLED TIME-SERIES CROSS-SECTION REGRESSION FOR 85 LARGE BANK HOLDING COMPANIES RELATING THE MARKET-TO-BOOK ASSET RATIO TO VARIOUS DETERMINANTS OF MARKET POWER 1971–86, FOURTH-QUARTER DATA (STANDARD ERRORS IN PARENTHESES)

$\bar{R}^2$	0.42
Number of Observations	1360
Dependent Variable Mean (Market-to-Book Asset Ratio, q)	1.00
Intercept	–0.94*** (0.011)
Branching Variables	
Liberalization of Branching Law	–0.0046*** (0.0017)
Liberalization of Multibank Holding Co. Law	–0.0074*** (0.0015)
Liberalization of Interstate Entry Law	0.00050 (0.0024)
Control Variables	
Dummy for Being on Compustat in 1964	0.0054*** (0.0013)
Dummy for Multinational Status	–0.0053** (0.0023)
Balance Sheet Variables	
Book-Value Asset Growth Since 1970	0.21*** (0.010)
Demand Deposits/Total Deposits ($\times 100$)	0.00056*** (0.000067)
Loans/Total Assets	0.0098 (0.012)
Foreign Deposits/Total Deposits ($\times 100$)	0.00019*** (0.000045)
Book Value of Assets	–4.09E ^{–9} (4.49E ^{–8})
(Cash and Treasury Securities)/Total Assets	–0.000027 (0.00013)
Financial Variables	
N.Y. Composite Index	0.0038*** (0.000030)
3-Month Treasury-Bill Rate	–0.00078** (0.00032)
20-Year Treasury-Bond Rate	–0.0019*** (0.00047)

*Significant at the 10-percent level.

**Significant at the 5-percent level.

***Significant at the 1-percent level.

The balance sheet variables generally have signs consistent with the notion that market power arises in deposit and loan markets, although statistically significant effects are found only for deposit markets. The fraction of demand and the fraction of foreign deposits in total deposits are positively and significantly related to the market-to-book asset ratio. The point estimate of the effect of the fraction of loans in total assets on the

market-to-book ratio is positive, and the estimate of the effect of the ratio of cash and treasury securities to total assets is negative, although neither variable is significant. Banks with more rapid asset growth in the past have significantly higher market-to-book ratios, perhaps because more rapid growth is associated with lack of competition or success due to other factors. Asset size per se, however, is not significant.

The control variable, whether a bank was on Compustat in 1964, is positively and significantly related to the market-to-book ratio, perhaps indicating that older, prominent bank holding companies that survived to be included in the sample have higher market values than bank holding companies that entered the sample later. The variable for multinational status is negatively related to the market-to-book ratio, which might be due to the very competitive international environment in which these 16 money center banks (as defined by the Federal Reserve) operate.

Finally, the effects of the financial variables are much as one might expect. Stock market values are positively related to the market-to-book ratio, and interest rates are negatively related.

Overall, the high correspondence between the expected effects of the variables and their estimated effects suggests that the market-to-book ratio is in fact a proxy for market power. Next, I test whether this proxy for market power is negatively related to bank risk taking.

E. *Bank Risk*

Below, the effects of the market-to-book asset ratio on two measures of bank default risk are examined. As discussed above, a key hypothesis is that the decline in banks' market power, as proxied by their market-to-book ratios, was a primary cause of the decline in banks' capital-to-asset ratios. Moreover, the theory implies that the cross-sectional variation in bank capital ratios would be influenced by variations in market power—banks with greater market power should have higher capital ratios.

F. *Market-Value Capital-to-Asset Ratios*

Table 3 presents coefficient estimates of equation (15), in which the market-value capital-to-asset ratio (i.e., the market value of capital divided by the market value of assets, defined as the market value of equity divided by the market value of equity plus the book value of liabilities) is regressed on the market-to-book asset ratio, q , holding

stock market and interest rate trends constant. It is necessary to hold stock market and interest rate trends constant since these trends potentially could influence both the dependent and independent variables, thus leading to spurious correlation. In addition, dummies for being on Compustat in 1964 and for multinational status are included.

In the first column of Table 3, ordinary least squares (OLS) estimates from a pooled time-series cross-section regression are reported. The OLS results suggest a strong, positive, and statistically significant relationship between the proxy for market power, the market-to-book asset ratio, and the market-value capital-to-asset ratio. Thus, as predicted, banks with more market power appear to hold more capital relative to assets. Moreover, the estimated magnitude of the effect is large, with a 10 percentage point increment in the market-to-book asset ratio leading to a 0.09 increase in the market-value capital-to-asset ratio, and is not sensitive to whether the variation in the market-to-book ratio is due to changes over time or differences across banks.¹³

There are, however, several reasons why the OLS estimates should be viewed with caution. First, endogeneity between q and bank risk is possible. For example, a bank

¹³To assess how sensitive the results were to pooling the cross sections over time, separate cross section regressions were run for each year from 1971 to 1986. Each of the OLS point estimates of the effect of the market-to-book asset ratio on the market-value capital-to-asset ratio were significantly different from zero at the 1-percent level and ranged from 0.62 to 1.19, approximately the same magnitude as the pooled time-series cross section results reported in Table 3.

Since different banks can have different responses to the market index (as proxied by the New York Composite), I also estimated an unconstrained version of equation (15) with separate intercepts and separate slope coefficients for the New York Composite Index for all 85 bank holding companies. However, the estimate of the effect of the market-to-book ratio was statistically significant and about the same magnitude as in the constrained model estimates reported in Table 3. Thus, the results appear robust regarding the source of variation in the market-to-book capital ratio—both the time-series and cross-sectional variation in banks' market-value capital-to-asset ratios are positively associated with time-series and cross-sectional variation in banks' market-to-book capital ratios.

TABLE 3—POOLED TIME-SERIES CROSS SECTION REGRESSION RELATING THE MARKET-VALUE CAPITAL-TO-ASSET RATIO TO THE MARKET-TO-BOOK ASSET RATIO AS A PROXY FOR MARKET POWER, FOURTH-QUARTER DATA, 1971–86
(STANDARD ERRORS IN PARENTHESES)

	OLS	TSLS ^a	TSLS ^b
$\bar{R}^2$	0.83	0.66	0.15
Number of Observations	1360	1360	1360
Dependent Variable Mean (Market-Value Capital-to-Asset Ratio)	0.054	0.054	0.054
Intercept	−0.85*** (0.014)	−0.70** (0.026)	−1.00*** (0.29)
N.Y. Composite Index	−0.0000044 (0.000013)	0.000032** (0.000015)	−0.000052 (0.000029)
3-Month Treasury-Bill Rate	−0.00042** (0.00018)	−0.00043** (0.00018)	−0.00042 (0.00051)
20-Year Treasury-Note Rate	−0.00023 (0.00025)	−0.00091*** (0.00028)	0.00065 (0.0015)
Dummy for Being on Compustat in 1965	0.0018** (0.00073)	0.0021 (0.00076)	−0.054 (0.046)
Dummy for Multinational Status	−0.018*** (0.00094)	−0.018*** (0.00098)	−0.032** (0.015)
Market-to-Book Asset Ratio (q)	0.91*** (0.013)	0.77*** (0.025)	1.09*** (0.27)

*Significant at the 10-percent level.

**Significant at the 5-percent level.

***Significant at the 1-percent level.

^aIncludes as instruments all variables on right-hand side of regression reported in Table 2.

^bIncludes as instruments only the branching, multibank holding company, and interstate expansion dummies, the financial variables, the on-Compustat dummy, and the dummy for multinational status.

with greater default risk could have a greater market-to-book asset ratio if deposit insurance were underpriced and its value were capitalized in a bank's market (but not book) value. Second, the market-to-book asset ratio measures market power with error due to *ex post* asset return realizations that are different from *ex ante* expectations. Third, q and the market-value capital-to-asset ratio might be spuriously correlated due to the presence of the market value of the bank's equity on both sides of the equation. Although the equation is not an identity, to the extent the ratio of the book value of liabilities to the book value of assets were approximately constant or much less variable than the ratio of the market value of equity to the book value assets, the estimated OLS coefficient on q would be biased toward 1. For these reasons, a simultaneous equations model is employed, and two-stage least squares (TSLS) estimates also are displayed in Table 3. By employing

TSLS techniques, exogenous variation in q is related to actual variation in market-value capital-to-asset ratios thus avoiding the potential endogeneity and measurement error problems associated with q . TSLS techniques also solve the problem of potential spurious correlation since the actual market value of common equity is not a right-hand-side variable. (The method of estimation employed produces standard errors corrected for the two-step nature of the procedure.)

In the first-column TSLS estimates, the instruments used to predict the market-to-book capital ratio include all of the explanatory variables in equation (14). In the second-column TSLS estimates, only the branching, multibank holding company and interstate expansion dummies, the financial variables, the on-Compustat dummy, and the multinational dummy were included as instruments, variables believed to be exogenous. Thus, in these second-column esti-

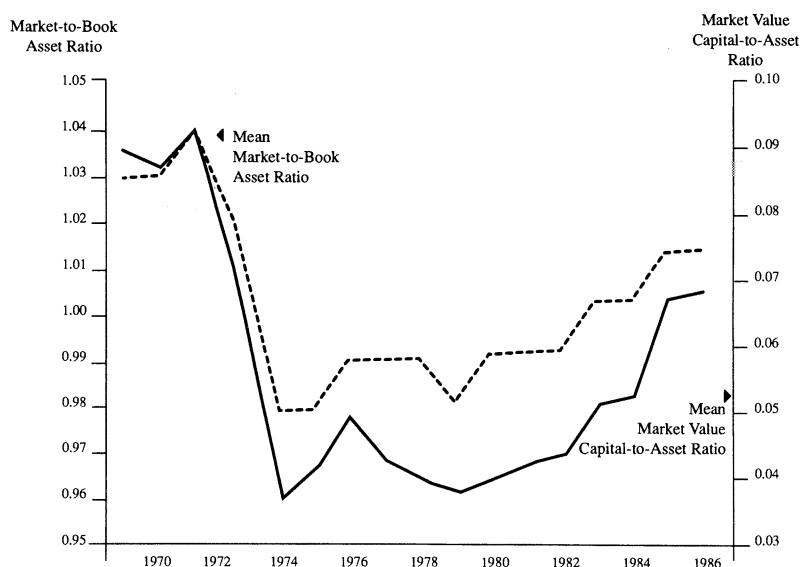


CHART 3. MEAN MARKET-TO-BOOK ASSET RATIO AND MARKET-VALUE CAPITAL-TO-ASSET RATIO

mates, the exogenous variation in q is due mainly to exogenous variation in the branching variables. However, the TSLS estimates of the effect of the market-to-book capital ratio on the market-value capital-to-asset ratio are very similar to the OLS estimate, especially the second TSLS estimate. This may be because the biases due to measurement error and endogeneity due to the capitalization of underpriced deposit insurance are offsetting. Thus, the finding that banks with more market power hold more capital relative to assets is robust with respect to the estimation method and specification of the model.

Moreover, the decline over time in banks' market-value capital-to-asset ratios is strongly associated with the decline in their market-to-book asset ratios. Chart 3 shows that the mean of banks' market-value capital-to-asset ratios follows the mean of their market-to-book ratios over time. According to the theory developed in Furlong and Keeley (1987, 1989), banks with more capital have less incentive to increase asset risk. Thus, as long as the stringency of asset risk regulation is not less at banks with stronger capital positions, such banks should have

lower default rates and thus represent lower risk exposures to the FDIC.¹⁴ In the next section, I present a test of this hypothesis, proxying risk exposure with the interest cost on large, uninsured CD's.

G. Interest Cost of Large CD's

Ideally, one would like to relate market power in banking directly to the risk exposure of the FDIC and a bank's uninsured depositors. While the FDIC's risk exposure is not directly observable,¹⁵ one can obtain a

¹⁴As Furlong and Keeley (1987a, 1989) show, the incentive to increase asset risk declines as leverage declines because the second derivative of the value of the insurance put option with respect to asset risk with respect to leverage is positive. This result holds in a multistate state preference or a continuous option pricing framework, but it does not hold for the simple two-state model presented in this paper.

¹⁵As Marcus and Shaked (1984) and Ronn and Verma (1986) have shown, it is possible to estimate the FDIC's risk exposure by using an options pricing model that relates the observed risk of bank equity (with deposit insurance) to the unobserved risk of bank assets (absent deposit insurance). However, such estimates require assumptions about bank closure policy, which bank and bank holding company liabilities are

measure of the risk premium on a bank's uninsured deposits, assuming that uninsured depositors behave as if they are not implicitly fully insured. The hypothesis is that the rate on large (over \$100,000) certificates of deposit (CD's) contain a risk premium related to the bank's default risk, which should be negatively related to a bank's market power as reflected in its market-to-book asset ratio, q .

Although there is a debate regarding whether, in fact, large CD's are implicitly fully insured (and thus whether they contain a risk premium), recent empirical evidence in Timothy H. Hannan and Gerald Hanweck (1988) and Christopher James (1987) strongly suggests that they do contain a risk premium. Moreover, if large CD's are sold in a national (or international) market, it is hard to imagine other factors that could explain the rate differences among banks on CD's of identical maturity.

The interest cost of large CD's is estimated from information contained in the Bank Consolidated Report of Condition and Income (the Bank Call Report). The average rate on large CD's is estimated by dividing the total interest paid (by all of the banks in the bank holding company) by the average dollar value outstanding during the year. Because of difficulties in constructing consolidated interest costs and amounts outstanding for all of the 85 bank holding companies in the previous sample, the sample had to be restricted to 77 bank holding companies for which complete information could be obtained.

Following James (1987), a weighted average maturity of CD's also is constructed to control for differences in the maturities of CD's outstanding. Since the data needed to construct this variable have only been collected since 1984, only the 1984–86 data are

used. Following James, I also control for time-series variation in the level of interest rates by including the average yield on three-month Treasury bills in the CD rate regressions.

The regression results are reported in Table 4. In the first column, estimates of the effect of the market-value capital-to-asset ratio on the CD rates are presented. As mentioned above, banks with more market power, as reflected in higher market-to-book asset ratios, hold more capital relative to assets, in theory, in order to protect their valuable charters. If this theory is correct, banks with more capital relative to assets should have lower default probabilities and thus should have lower CD rates. The results in the first column of Table 4 confirm this hypothesis: banks with greater market-value capital-to-asset ratios pay lower CD rates. In fact, a 1 percentage point increase in a bank's capital-to-asset ratio would lower its CD rate by 14 basis points.¹⁶ This result is somewhat larger but comparable to the 8-basis-point effect Hannan and Hanweck (1988) found using the book-value capital-to-asset rates in a somewhat different specification.

In the second and third columns of Table 4, estimates of the effects of the market-to-book asset ratio on the CD rate are presented. As in the previous equations reported in Table 3, both OLS and TSLS results are presented. The TSLS results are obtained by using all of the right-hand-side variables in equation (14) (less one interest rate trend variable, due to degrees of freedom limitations) as instruments. The results confirm the hypothesis that banks with more market power have lower default risk.

As expected, the coefficient on the market-to-book asset ratio is negative and significantly different from zero at the 1-percent level. Moreover, the effect, while not large, is economically meaningful (each 1 percentage point increment in the market-to-book asset ratio reduces the average CD

insured, and a number of other factors (such as the market value of preferred stock, whether bank holding company assets will be used to support a failing bank, etc.). Because of uncertainty regarding just what assumptions might be appropriate and possible changes in a number of these factors over time, I focus on CD rates, a more directly observable proxy for bank default risk.

¹⁶This result is also consistent with the hypothesis that banks with larger capital-to-asset ratios have less incentive to increase asset risk.

TABLE 4—POOLED TIME-SERIES CROSS SECTION REGRESSION RELATING INTEREST COST OF LARGE CD'S TO THE MARKET-TO-BOOK ASSET RATIO AS A PROXY FOR MARKET POWER AND THE MARKET-VALUE CAPITAL-TO-ASSET RATIO, FOURTH-QUARTER DATA, 1984–86 (77 BANK HOLDING COMPANIES)
(STANDARD ERRORS IN PARENTHESES)

	OLS	OLS	TOLS
$\bar{R}^2$	0.43	0.43	0.39
Number of Observations	231	231	231
Dependent Variable Mean (Interest Cost of CD's Divided by CD's Outstanding)	0.085	0.085	0.085
Intercept	0.029*** (0.0064)	0.020*** (0.0043)	0.19*** (0.070)
3-Month Treasury-Bill Rate	0.83*** (0.074)	0.82*** (0.074)	0.81*** (0.084)
Average Maturity of CD's	0.0011*** (0.00025)	0.0011*** (0.00025)	0.00073 (0.00055)
Market-to-Book Asset Ratio (q)		-0.18*** (0.042)	-0.16** (0.066)
Market-Value Capital-to-Asset Ratio	-0.14*** (0.034)		

*Significant at the 10-percent level.

**Significant at the 5-percent level.

***Significant at the 1-percent level.

cost by 16–18 basis points). It is important to recognize that this estimated effect arises from both the time-series as well as the cross-sectional variation in CD costs. This result is consistent with the significant positive effect of the market-to-book ratio on the market-value capital-to-asset ratio reported in Table 3 and the significant negative relationship between the market-value capital-to-asset ratio and the CD risk premium reported in column one of Table 4.

III. Summary and Conclusion

This paper addresses two major empirical puzzles. Why has the deposit insurance system worked as well as it has over much of its history even though it provides a moral hazard for excessive risk taking, and why is the cross-sectional distribution of bank risk taking nonuniform? The hypothesis is that various anticompetitive restrictions endowed banks with market power and made banking charters valuable. The potential loss of a charter in the event of bankruptcy created, in effect, a regulatory bankruptcy cost, which counterbalanced the incentive

for excessive risk taking due to fixed-rate deposit insurance.

The empirical results are consistent with this hypothesis. Banks with more market power, as reflected in larger market-to-book asset ratios, hold more capital relative to assets (on a market-value basis) and they have a lower default risk as reflected in lower risk premiums on large, uninsured CD's. Thus, at least some of the increase in bank and thrift failures and payouts from the deposit insurance funds may be due to a general decline in the value of bank charters associated with increased competition within the banking and financial service industry.

In the past, the perverse incentives created by the deposit insurance system were countervailed by the potential loss of a valuable charter that induced banks to limit their own risk taking. This does not mean that it is desirable or even possible to return to a system of anticompetitive restrictions in order to reduce banking risk. But it does mean that the deposit insurance system must be reformed to reduce the rewards it provides for excessive risk taking.

REFERENCES

- Amel, Dean F., and Keane, Daniel G., "State Laws Affecting Commercial Bank Branching, Multibank Holding Companies, and Interstate Banking," Working Paper, Board of Governors of the Federal Reserve System, 1987.
- Benston, George J., "The Optimal Banking Structure," *Journal of Bank Research*, Winter 1973, 3, 220-37.
- Berger, Allen N. and Hannan, Timothy H. "The Price Concentration Relationship in Banking," Working Paper No. 100, Board of Governors of the Federal Reserve System, June 1987.
- Dothan, Uri and Williams, Joseph, "Banks, Bankruptcy, and Public Regulation," *Journal of Banking and Finance*, March 1980, 4, 65-87.
- Flannery, Mark J., "The Social Costs of Unit Banking Restrictions," *Journal of Monetary Economics*, March 1984, 13, 237-49.
- Furlong, Frederick T. and Keeley, Michael C., "Bank Capital Regulation and Asset Risk," *Economic Review*, Federal Reserve Bank of San Francisco, Spring 1987, 20-40.
- , "Bank Capital Regulation and Risk Taking: A Note," *Journal of Banking and Finance*, November 1989, 13, 883-91.
- Hannan, Timothy H. and Hanweck, Gerald, "Bank Insolvency Risk and the Market for Large CDs," *Journal of Money, Credit, and Banking*, May 1988, 20, 438-46.
- James, Christopher, "Off-Balance-Sheet Banking," *Economic Review*, Federal Reserve Bank of San Francisco, Fall 1987, 21-36.
- Kareken, John H. and Wallace, Neil, "Deposit Insurance and Bank Regulation: A Partial Equilibrium Exposition," *Journal of Business*, July 1978, 51, 413-38.
- Keeley, Michael C., (1985a) "The Regulation of Bank Entry," *Economic Review*, Federal Reserve Bank of San Francisco, Summer 1985, 5-13.
- , (1985b) "Bank Entry and Deregulation," *Weekly Letter*, Federal Reserve Bank of San Francisco, August 25, 1985.
- , "Bank Capital Regulation in the 1980s: Effective or Ineffective," *Economic Review*, Federal Reserve Bank of San Francisco, Winter 1988, 1-20.
- Kling, Arnold, "The Banking Crisis from a Macroeconomic Perspective," Working Paper, Board of Governors of the Federal Reserve System, 1986.
- Lindenberg, Eric and Ross, Stephen, "Tobin's q Ratio and Industrial Organization," *Journal of Business*, January 1981, 54, 1-32.
- Marcus, Alan J., "Deregulation and Bank Financial Policy," *Journal of Banking and Finance*, December 1984, 8, 557-65.
- and Shaked, Israel, "The Valuation of FDIC Deposit Insurance Using Option-Pricing Estimates," *The Journal of Money, Credit, and Banking*, November 1984, Part 1, 16, 446-60.
- Merton, Robert C., "An Analytic Derivation of the Cost of Deposit Insurance Loan Guarantees," *Journal of Banking and Finance*, June 1977, 1, 3-11.
- Peltzman, Sam, "Entry into Commercial Banking," *Journal of Law and Economics*, October 1965, 8, 11-50.
- Pennacchi, George, "A Reexamination of the Over- (or Under-) Pricing of Deposit Insurance," *Journal of Money, Credit, and Banking*, August 1987, 19, 340-60.
- Ronn, Ehud and Verma, Avinash K., "Pricing Risk-Adjusted Deposit Insurance: An Option-Based Model," *Journal of Finance*, September 1986, 41, 871-94.
- Salinger, Michael, "Tobin's q , Unionization, and the Concentration-Profits Relationship," *Rand Journal of Economics*, Summer 1984, 15, 159-70.
- Savage, Donald T. and Rhoades, Stephen A., "The Effect of Branch Banking on Pricing, Profits, and Efficiency of Unit Banks," in *Proceedings of a Conference on Bank Structure and Competition*, Federal Reserve Bank of Chicago, 1979.
- and Solomon, Elinor H., "Branch Banking: The Competitive Issues," *Journal of Bank Research*, Summer 1980, 11, 110-21.
- Sharpe, William F., "Bank Capital Adequacy, Deposit Insurance, and Security Values," *Journal of Financial and Quantitative*

- Analysis Proceedings*, November 1978, 13, 701–18.
- Smirlock, Michael, Gilligan, Thomas and Marshall, William**, “Tobin’s q and the Structure Performance Relationship,” *American Economic Review*, December 1984, 74, 1051–60.
- Stigler, George**, “A Theory of Oligopoly,” *Journal of Political Economy*, February 1964, 72, 44–61.